Examiner only

5. *(a)* Describe in detail how you would determine the wavelength of light from a (low-power) laser, given an opaque plate with two parallel slits. (The distance between the centres of the slits is less than 1 mm and has already been measured.) [6 QER]

(b) A diffraction grating is used with a laser as shown, and a line of bright spots is seen on the screen as shown.

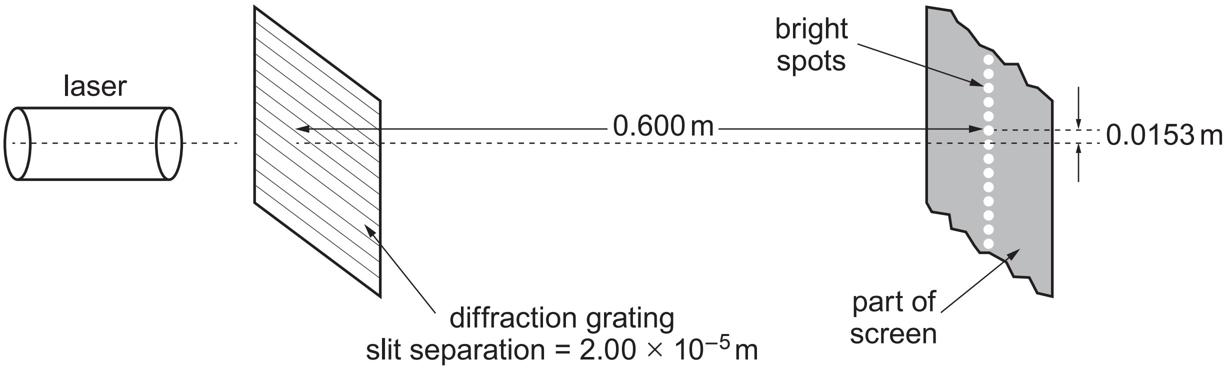

DIAGRAM NOT TO SCALE

(i) Using a labelled sketch of a relevant triangle (in the space below), explain why, to a very good approximation:

$$\sin\theta_1 = \frac{0.0153}{0.600}$$

in which θ_1 is the angle between the zeroth order and first order beams. [2]

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(ii) Determine the wavelength of light from the laser. [2]

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Examiner only

(iii) Calculate the highest **order** produced by this grating for this wavelength, and suggest why so many orders are produced by this grating. [2]

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